SUNJETA JAHAN SNEGDHA

Computer Science & Engineering Undergraduate | Full-Stack Developer | AI/ML Researcher

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PROFILE SUMMARY

Computer Science and Engineering undergraduate at AIUB (CGPA 3.87/4.00) focused on backend and full-stack development: REST APIs and web applications with NestJS, Next.js, TypeScript and PostgreSQL, and PHP/MySQL web applications, with additional projects in C#, Java and C++. Author of 3 IEEE conference papers on explainable machine learning, first author on one.

EDUCATION

American International University-Bangladesh (AIUB) Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering | Major: Software Engineering **CGPA: 3.87 / 4.00**

Relevant Coursework: Advanced Web Technology, OOP, Data Structures & Algorithms, Database Systems, Software Requirements Engineering, Software Quality & Testing, Machine Learning

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP, C++, SQL | Familiar: C#, Java, Python

Backend: NestJS, Node.js, REST APIs, JWT Authentication, PHP (MVC)

Frontend: Next.js, React, Tailwind CSS, HTML, CSS, AJAX

Databases: PostgreSQL, MySQL, Microsoft SQL Server, TypeORM

Core Concepts: OOP, Data Structures & Algorithms, MVC, Role-Based Access Control, Relational Database Design

ML & Research: Random Forest, XGBoost, SVM, SMOTE, SHAP, Probability Calibration

Tools & Desktop: Git, GitHub, VS Code, Visual Studio, LaTeX, .NET (Windows Forms), OpenGL / GLUT

FEATURED PROJECTS

PawsHome BD | Multi-Role Pet Adoption Platform *NestJS, Next.js, TypeScript, TypeORM, PostgreSQL* GitHub

- Team platform where listers post pets, admins verify and approve them, adopters adopt and vets provide care.
- NestJS REST API with JWT authentication (custom JwtAuthGuard), DTO validation, mailer and image uploads (Multer).
- PostgreSQL data model with one-to-one and one-to-many TypeORM relations; Next.js frontend.

Travel Management System *PHP (MVC), MySQL, JavaScript (AJAX)* GitHub

- Scouts submit destinations, admins approve, users browse with live AJAX search, wishlists, comments and trip-cost estimates.
- Secured with CSRF protection, prepared statements on every query and role-based login with Remember Me.

Cafe La Vien | Cafe Management System *C#, .NET Windows Forms, SQL Server* GitHub

- Desktop app with Admin (users, menu, sales) and Cashier (orders, cart, invoices) workflows, VAT billing and a shared data-access layer.

Bank Management System *Java, OOP*

- Deposits, withdrawals and inter-account transfers with a transaction history of IDs, dates, amounts, senders and receivers.

A Day in Winter: Rural Village Life *C++, OpenGL, GLUT* GitHub

- Four animated scenes (~2,000 lines) from modular OpenGL drawing functions, with timer animation and keyboard/menu control.

RESEARCH & PUBLICATIONS

Machine Learning Research | *Undergraduate Researcher, Department of Computer Science, AIUB*

- **Phishing detection:** 7 classifiers on 1,082,664 URLs from 8 sources; Random Forest best, SHAP explanations, McNemar's test.
- **Stroke risk:** LR, RF, XGBoost and SVM on imbalanced clinical data (SMOTE), with SHAP and Platt / isotonic calibration.
- **Agentic AI:** built a predictive framework for balancing autonomy and human oversight in agentic AI adoption.

Publications

1. T. A. Fahim, F. Mahmud, **S. J. Snegdha**, R. A. M. Rudro, T. Sayeed, and K. Nur, "Phishing URL Detection Using Machine Learning: A Comparative Study with SHAP Based Explainability on a Large-Scale Multi-Source Dataset," *IEEE ICACT 2026*, Kathmandu, Nepal.
2. **S. J. Snegdha**, T. A. Fahim, R. A. M. Rudro, T. Sayeed, and K. Nur, "Balancing Autonomy and Oversight: A Predictive Framework for Agentic AI Adoption," *IEEE ICACT 2026*, Kathmandu, Nepal.
3. T. A. Fahim, **S. J. Snegdha**, T. Rahannum, R. A. M. Rudro, and K. Nur, "Explainability and Calibration of Stroke Risk Prediction Models Based On AI," *IEEE COMPAS 2026*, University of Dhaka, Bangladesh (accepted).

ACHIEVEMENTS

- **Finalist,** Three Minute Thesis (3MT) Competition and Sciblitz 2.0 Science & Tech Contest, Chittagong University of Engineering & Technology (CUET)